

PREPRINTS

7 papers

An overview of the papers released at github.com/suvrit/preprints. Except where marked, the mathematics in these papers was developed in sustained dialogue with an AI assistant (GPT Pro); none of the results is zero-shot. Claude Code was used extensively for writing, editing, and repository management. Results are grouped by subject; entry numbers do not indicate a ranking. Each summary is followed by the paper's title, linked to its latest timestamped PDF, together with its version and, where one exists, its arXiv identifier.

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COMBINATORICS

1. McLeod's 1959 conjecture. The paper proves McLeod's conjecture that $(h_{\ell+r}/h_\ell)^{1/r}$ is subadditive for all $\ell \geq 0$ and $r \geq 1$, where h_k is the complete homogeneous symmetric polynomial of degree k . The first step shows, via a closed queueing network, that h_k/h_{k-1} is convex; a tensorization argument then couples neighboring ratios tightly enough to make their geometric mean convex.

A Proof of McLeod's 1959 Conjecture on Complete Homogeneous Symmetric Ratios · v1, 5 October 2026 · arXiv:2610.07302

2. Monotone Turán-ratios of Schur polynomials. The paper proves that, unless all n padded parts of λ are equal, the Turán-ratio $S_{\lambda,i}(x)^2/(S_{\lambda,i-1}(x)S_{\lambda,i+1}(x))$ is strictly increasing for $x > 1$ and $1 \leq i < n$, where $S_{\lambda,i}(x) = s_\lambda(1^{[n-i]}, x^{[i]})$ is the Schur polynomial in n variables with i arguments equal to x and the rest equal to 1. For one-row shapes this answers a question of Ait-Haddou on complete homogeneous symmetric polynomials; in that case a second proof, through the zeros of Meixner polynomials, gives an explicit positive formula for the logarithmic derivative of the ratio. It conjectures the analogue for Macdonald polynomials.

Monotonicity of Turán-ratios of symmetric polynomials · v1, 7 October 2026

3. A Kotljanskii inequality for permanents. The paper proves that $\text{per}(A_{S \cup T}) \text{per}(A_{S \cap T}) \geq \text{per}(A_S) \text{per}(A_T)$ for all $S, T \subseteq [n]$ when A is an inverse M -matrix that becomes symmetric after a positive diagonal similarity; here A_S is the principal submatrix indexed by S . The proof writes permanents as moments of a complex Gaussian vector and applies Ginibre's correlation inequality. An explicit counterexample shows that symmetry cannot be dropped.

A Kotljanskii inequality for permanents · v1, 2 October 2026 · arXiv:2609.39979

4. Majorization for normalized Jack and Macdonald polynomials. The paper proves that if λ majorizes μ , then the Jack polynomial P_λ in n variables, normalized to equal one at $(1, \dots, 1)$, is at least the normalized P_μ on the positive orthant, for every n , and strictly so when the coordinates are distinct. For the conjectured Macdonald analogue, it proves the rank-two case, the case of a one-row partition against the hook below it for $t \leq q$ in every rank, and, by exact certificates, the full statement in three variables for partitions of size at most eight. It shows that the Harish-Chandra–Itzykson–Zuber mechanism behind the Schur and Jack proofs has no direct finite- q analogue. The full conjecture remains open.

On majorization inequalities for normalized Jack and Macdonald polynomials · v1, 8 October 2026

MATRIX ANALYSIS

5. Lin's singular-value conjecture. The paper proves Lin's conjecture that $\lambda_j((1-t)A + tB) \geq \sigma_j^{1/2}(A^{2(1-t)}B^{2t})$ for all positive semidefinite A, B , all j , and every $t \in [0, 1]$. The case $t = \frac{1}{2}$ is Drury's theorem, which answered a question of Bhatia and Kittaneh. The proof follows Drury's argument but replaces its block-matrix step, which works only at the midpoint, with an eigenvalue count based on an inertia identity and a one-variable inequality.

On Lin's Arithmetic–Geometric Means Conjecture for Singular Values · v1, 8 October 2026

6. Bellman's 1959 claim on Hua–Bellman matrices. The paper shows that Bellman's 1959 strengthening of Hua's results on positive definiteness of the matrices $[\det(I - A_i^* A_j)^{-\alpha}]_{ij}$, for strict contractions A_i , rests on an erroneous proof and, by an explicit example, that it is false. It gives conditions under which $\det(I - A^* B)^{-\alpha}$ is a positive definite function: for complex contractions, the known sharp condition, with the integer exponents below Hua's range reached by an elementary argument; for real symmetric contractions, the part of Bellman's claim that survives. It also introduces a Poincaré-like distance on noncommuting strict contractions.

Positive definite functions of noncommuting contractions, Hua–Bellman matrices, and a new distance metric · v1, 2 October 2026 · arXiv:2112.00056v2 · Not AI-assisted

COMPLEX GEOMETRY

7. Schur forms under decomposable curvature. Griffiths asked in 1969 whether positive curvature of a Hermitian holomorphic vector bundle forces positivity of its characteristic forms; Du's rank-nine example shows that a strictly Griffiths-positive metric can have a top Chern form that is negative on an open set. The paper proves that every Schur form is weakly nonnegative wherever the curvature is decomposable, that is, a sum of a completely positive map and a completely copositive map, with no splitting assumption. The algebraic input is the inequality $\Phi_n(H) \geq 0$ for Finski's double mixed discriminant of every decomposable map H , obtained from a quaternionic Gaussian determinant formula. Beyond the decomposable cone, $\Phi_k \geq 0$ on every square compression of H is equivalent to $\Phi_n(H + S) \geq 0$ for every entanglement-breaking map S ; the class of maps satisfying this condition contains indecomposable maps but is not convex for $n \geq 9$, since Du's map is an average of four of its members.

Weak positivity of Schur forms under decomposable curvature · v1, 8 October 2026

